Scorecard (procreview.jsp?sid=aaaN5tjtX0b7WgArBjowySun Jan 08 23:44:45 IST 2023&qsetId=tME wLqNOjk=&qsetName=)

Accuracy (AccSelectGraph.jsp?sid=aaaN5tjtX0b7WgArBjowySun Jan 08 23:44:45 IST 2023&qsetId=tME wLqNOjk=&qsetName=)

Qs Analysis (QsAnalysis.jsp?sid=aaaN5tjtX0b7WgArBjowySun Jan 08 23:44:45 IST 2023&qsetId=tME wLqNOjk=&qsetName=)

Video Attempt / Solution (VideoAnalysis.jsp?sid=aaaN5tjtX0b7WgArBjowySun Jan 08 23:44:45 IST 2023&qsetId=tME wLqNOjk=&qsetName=)

Solutions (Solution.jsp?sid=aaaN5tjtX0b7WgArBjowySun Jan 08 23:44:45 IST 2023&qsetId=tME wLqNOjk=&qsetName=)

Bookmarks (Bookmarks.jsp?sid=aaaN5tjtX0b7WgArBjowySun Jan 08 23:44:45 IST 2023&qsetId=tME wLqNOjk=&qsetName=)

Section-1

Sec 1

Directions for questions 1 to 4: Answer the questions on the basis of the information given below.

A famous detective, Byomkesh, was solving a problem when he accidently spilled water on the sheet of paper. Byomkesh challenged Ajit to solve the calculation by replacing the digits. Here is the calculation:

(i)
$$\begin{array}{r} ABCD \\ \times \quad E \\ \hline =FCDIE \end{array}$$

(ii)
$$\begin{array}{r} JEAG \\ \times \quad C \\ \hline =BBFHH \end{array}$$

Here, all the letters (A to J) represent distinct single digit non-negative integers from 0 to 9. It is also known that C and D are even natural numbers and I is an odd natural number.

Q.1 [11831809]
What is the value of A?

Solution:
Correct Answer : 7

Answer key/Solution

Step 1:
Let's begin with the first equation. You should remember that all digits are distinct.

- D × E has a unit digit equal to E. D is an even natural number, so D × E must be even too, therefore E must be an even number. If you put D = 2, and then E equal to 4, 6 or 8, you will not get E back at unit's digit. The same way, D = 4 or 8 wont given you a proper result. The equation only works if **D = 6.** E can therefore be 2, 4 or 8.
  Considering E = 2, gives us 12, ensuring we carry over 1. Thereafter, E × C (an even number) + 1 (carry over) = Odd number, which is what I is.
  No other value of E (4 or 8) will give a carry-over that is odd. 6(D) × 4 carries over 2, while 6 × 8 carries over 4. In the last two cases, I will come out to be an even number violating the basic conditions given. Thus, **E = 2**. Therefore, C is either 4 or 8, and I is either 9 or 7.
- If C = 8, then E(2) × 8 carries over 1. A carry-over of 1 does not work because E(2) × B + carry over = 6. Since E × B is even, the carry-over has to be even too to get D = 6 as unit digit. **So C = 4, and I = 9. Thus, B = 3.**
- Thereafter, E(= 2) × A can only have a unit digit of 4 (C) if **A = 7.**

Step 2:

- Having figured out all letters used in the first equation, the second equation has G × 4 giving H, therefore H has to be an even digit. One possible option is 8, for which G would have to be 2, which is not possible as E = 2. Therefore, **H has to be 0 and G has to be 5.**
- Continuing with the equation, **J will come out to be 8.**

The value of A is 7.

Bookmark FeedBack

Directions for questions 1 to 4: Answer the questions on the basis of the information given below.

A famous detective, Byomkesh, was solving a problem when he accidently spilled water on the sheet of paper. Byomkesh challenged Ajit to solve the calculation by replacing the digits. Here is the calculation:

(i)
$$\begin{array}{r} ABCD \\ \times \quad E \\ \hline =FCDIE \end{array}$$

(ii)
$$\begin{array}{r} JEAG \\ \times \quad C \\ \hline =BBFHH \end{array}$$

Here, all the letters (A to J) represent distinct single digit non-negative integers from 0 to 9. It is also known that C and D are even natural numbers and I is an odd natural number.

Q.2 [11831809]
What is the value of H?

Solution:
Correct Answer : 0

Answer key/Solution

Step 1:
Let's begin with the first equation. You should remember that all digits are distinct.

- D × E has a unit digit equal to E. D is an even natural number, so D × E must be even too, therefore E must be an even number. If you put D = 2, and then E equal to 4, 6 or 8, you will not get E back at unit's digit. The same way, D = 4 or 8 wont given you a proper result. The equation only works if **D = 6.** E can therefore be 2, 4 or 8.
  Considering E = 2, gives us 12, ensuring we carry over 1. Thereafter, E × C (an even number) + 1 (carry over) = Odd number, which is what I is.
  No other value of E (4 or 8) will give a carry-over that is odd. 6(D) × 4 carries over 2, while 6 × 8 carries over 4. In the last two cases, I will come out to be an even number violating the basic conditions given. Thus, **E = 2**. Therefore, C is either 4 or 8, and I is either 9 or 7.
- If C = 8, then E(2) × 8 carries over 1. A carry-over of 1 does not work because E(2) × B + carry over = 6. Since E × B is even, the carry-over has to be even too to get D = 6 as unit digit. **So C = 4, and I = 9. Thus, B = 3.**
- Thereafter, E(= 2) × A can only have a unit digit of 4 (C) if **A = 7.**

Step 2:

- Having figured out all letters used in the first equation, the second equation has G × 4 giving H, therefore H has to be an even digit. One possible option is 8, for which G would have to be 2, which is not possible as E = 2. Therefore, **H has to be 0 and G has to be 5.**
- Continuing with the equation, **J will come out to be 8.**

The value of H = 0.

Bookmark FeedBack

Directions for questions 1 to 4: Answer the questions on the basis of the information given below.

A famous detective, Byomkesh, was solving a problem when he accidently spilled water on the sheet of paper. Byomkesh challenged Ajit to solve the calculation by replacing the digits. Here is the calculation:

(i)
$$\begin{array}{r} ABCD \\ \times \quad E \\ \hline =FCDIE \end{array}$$

(ii)
$$\begin{array}{r} JEAG \\ \times \quad C \\ \hline =BBFHH \end{array}$$

Here, all the letters (A to J) represent distinct single digit non-negative integers from 0 to 9. It is also known that C and D are even natural numbers and I is an odd natural number.

Q.3 [11831809]
If Ajit replaces one of the letters representing an even natural number in equation (i) with an odd natural number, then the value of the result/ product increases by 60. Which number did Ajit replace?

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:
Let's begin with the first equation. You should remember that all digits are distinct.

- D × E has a unit digit equal to E. D is an even natural number, so D × E must be even too, therefore E must be an even number. If you put D = 2, and then E equal to 4, 6 or 8, you will not get E back at unit's digit. The same way, D = 4 or 8 wont given you a proper result. The equation only works if **D = 6.** E can therefore be 2, 4 or 8.
  Considering E = 2, gives us 12, ensuring we carry over 1. Thereafter, E × C (an even number) + 1 (carry over) = Odd number, which is what I is.
  No other value of E (4 or 8) will give a carry-over that is odd. 6(D) × 4 carries over 2, while 6 × 8 carries over 4. In the last two cases, I will come out to be an even number violating the basic conditions given. Thus, **E = 2**. Therefore, C is either 4 or 8, and I is either 9 or 7.
- If C = 8, then E(2) × 8 carries over 1. A carry-over of 1 does not work because E(2) × B + carry over = 6. Since E × B is even, the carry-over has to be even too to get D = 6 as unit digit. **So C = 4, and I = 9. Thus, B = 3.**
- Thereafter, E(= 2) × A can only have a unit digit of 4 (C) if **A = 7.**

Step 2:

- Having figured out all letters used in the first equation, the second equation has G × 4 giving H, therefore H has to be an even digit. One possible option is 8, for which G would have to be 2, which is not possible as E = 2. Therefore, **H has to be 0 and G has to be 5.**
- Continuing with the equation, **J will come out to be 8.**

The final product (F C D I E) equals 14692. For it to increase by 60, would mean it would be equal to 14752. That would be possible if we replace C = 4, with the digit 7.

Bookmark FeedBack

Directions for questions 1 to 4: Answer the questions on the basis of the information given below.

A famous detective, Byomkesh, was solving a problem when he accidently spilled water on the sheet of paper. Byomkesh challenged Ajit to solve the calculation by replacing the digits. Here is the calculation:

(i)
$$\begin{array}{r} ABCD \\ \times \quad E \\ \hline =FCDIE \end{array}$$

(ii)
$$\begin{array}{r} JEAG \\ \times \quad C \\ \hline =BBFHH \end{array}$$

Here, all the letters (A to J) represent distinct single digit non-negative integers from 0 to 9. It is also known that C and D are even natural numbers and I is an odd natural number.

Q.4 [11831809]
By replacing one of the digits, from J, E, A, and G, to 1 in equation (ii), the final product changes to a number that has four even digits, with the solitary one odd digit as its leading digit (at the ten thousand's place). Which of the following digits have been changed? The multiplier C has not been changed.

1 ◯ J

2 ◯ E

3 ◯ A

4 ◯ G

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
Let's begin with the first equation. You should remember that all digits are distinct.

- D × E has a unit digit equal to E. D is an even natural number, so D × E must be even too, therefore E must be an even number. If you put D = 2, and then E equal to 4, 6 or 8, you will not get E back at unit's digit. The same way, D = 4 or 8 wont given you a proper result. The equation only works if **D = 6.** E can therefore be 2, 4 or 8.
  Considering E = 2, gives us 12, ensuring we carry over 1. Thereafter, E × C (an even number) + 1 (carry over) = Odd number, which is what I is.
  No other value of E (4 or 8) will give a carry-over that is odd. 6(D) × 4 carries over 2, while 6 × 8 carries over 4. In the last two cases, I will come out to be an even number violating the basic conditions given. Thus, **E = 2**. Therefore, C is either 4 or 8, and I is either 9 or 7.
- If C = 8, then E(2) × 8 carries over 1. A carry-over of 1 does not work because E(2) × B + carry over = 6. Since E × B is even, the carry-over has to be even too to get D = 6 as unit digit. **So C = 4, and I = 9. Thus, B = 3.**
- Thereafter, E(= 2) × A can only have a unit digit of 4 (C) if **A = 7.**

Step 2:

- Having figured out all letters used in the first equation, the second equation has G × 4 giving H, therefore H has to be an even digit. One possible option is 8, for which G would have to be 2, which is not possible as E = 2. Therefore, **H has to be 0 and G has to be 5.**
- Continuing with the equation, **J will come out to be 8.**

If you change A (which equals 7) to 1, would find the product come out to be 32860, which satisfies the conditions laid down in the question.

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Seven friends - Arun, Bishwas, Chetan, Deepak, Eshan, Faiz and Gopal are subscribers to one of the Cellular Networks - Airtel, Jio, Voda and BSNL.
All members of every family subscribe to the network. Every Network has atleast one customer from among these 7 friends.
All the 7 friends are on a trip to Dubai on a short visit. Every one of made a call to their own families, and in addition atleast two calls to the families of their friends, who were accompanying them on the trip. The roaming charges per call of the 4 networks are : Airtel ($3) ; Jio ($4) ; Voda ($5) and BSNL ($6).
If a subscriber is calling from Network X and the receiver is from Network Y, the charges for the call is equal to Minimum of (Roaming Charges of Network X, Roaming Charges of Network Y).
If they are both from the same network, the charges for the call is equal to the roaming charges of that network.
The additional information are as below:
(i) The total call charges for two friends were $9 and $12. The friend(s) on Jio network called the family members of at least one friend, who was on the Airtel Network.
(ii) Faiz and Gopal talked to the family members of all friends and the average call charges between them was $30. The call charges for Faiz was more than that for Gopal, who is on the Voda network.
(iii) Deepak, who is on the BSNL network spend $16 on the call charges.
(iv) Arun called all the families who were receivers of calls from a friend who is a BSNL subscriber. All the families who received calls from Eshan belonged to different networks.
(v) No family received more than five calls. The family members of Bishwas and Chetan received 4 calls. Chetan had not called Eshan's family.
(vi) The total call charges of Deepak, Eshan and Bishwas were consecutive numbers in the decreasing order.
(vii) Family members of any friend can receive a call from atmost one Airtel subscriber.
(viii) Family members of two friends on the same network must be contacted by different number of friends.

Q.5 [11831809]
The family members who were on the Voda network received calls from

1 ◯ Arun, Bishwas, Eshan, Faiz and Gopal

2 ◯ Arun, Bishwas, Faiz and Gopal

3 ◯ Chetan, Eshan, Faiz and Gopal

4 ◯ Bishwas, Chetan, Eshan, Faiz and Gopal

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:

From condition (iii), Deepak, who is on the BSNL network spend $16 on the call charges.

From condition (vi), The total call charges of Deepak, Eshan and Bishwas were $16, $15 and $14.

From condition (ii), Faiz and Gopal talked to the family members of all friends and the average call charges between them was $30. So their total call charges were $60. The call charges for Faiz was more than that for Gopal, who is on the Voda network.

So, total call charges for Faiz were more than $30 and total call charges for Gopal were less than $30. Hence, Faiz is on the BSNL network.

Till now, it is clear that there are 2 on BSNL network and 1 of each i.e., Airtel, Jio and Voda networks.

Based on the above data; the total call charges for Faiz = 3 + 4 + 5 + 6 + 6 = $24.

And; that for Gopal = 3 + 4 + 5 + 5 + 5 = $22.

In order to satisfy; the average call charges for Faiz and Gopal together be equal to $30, there is only one possibility that charges for each of them were $3 and $4 for the remaining two calls.

So, these two calls were made to Airtel and Jio networks.

Therefore, Airtel, Jio, Voda and BSNL networks have 2, 2, 1 and 2 subscribers respectively.

Hence, total call charges for Faiz = 24 + 3 + 4 = $31 and total call charges for Gopal = 22 + 3 + 4 = $29.

Step 2:

From condition (i), The total call charges of $9 can be for Airtel subscriber only in the form of 3 + 3 + 3 (= $9).

The total call charges of $12 will also be applicable to the Airtel subscriber as the Jio subscriber will have to make calls to the family members of at least one friend on the Airtel network.

So, the call charges of Jio subscriber must be $4 + $3 for two mandatory calls i.e., one call to his family members and other to Airtel subscriber family members.

Next; Call charges for Jio subscribers can be either $3 or $4; A total call charges of $12 is not possible.

From condition (iv), The total call charges for Arun is $12 and that for Chetan is $9.

From the given information, Every one of made a call to their own families.

From condition (v), Chetan had not called Eshan's family.

The information can be shown in the following table.

Family Members	Friend						
	$12, Arun (Airtel)	$14, Bishwas (Jio)	$9, Chetan (Aitel)	$16, Deepak (BSNL)	$15, Eshan (Jio)	$31, Faiz (BSNAL)	$29,Gopal (Voda)
Arun ((Airtel)	3					3	3
Bishwas (Jio)		4				4	4
Chetan (Airtel)			3			3	3
Deepak (BSNL)				6		6	5
Eshan (Jio)			X		4	4	4
Faiz (BSNL)						6	5
Gopal (Voda)						5	5

Step 3:

From condition (iv), Arun called all the families who were receivers of calls from a friend who is a BSNL subscriber. All the families who received calls from Eshan belonged to different networks.

The total call charges for Bishwas are $14, which can be as 4 + 3 + 4 + 3. It means he must call to family members of both Airtel subscribers.

From condition (v), The family members of Bishwas and Chetan received 4 calls.

The total call charges for Deepak are $16 as 6 + 4 + 6 only.

Hence, the total information can be shown in the following table.

Family Members	Friend							Family Calls
	$12, Arun (Airtel)	$14, Bishwas (Jio)	$9, Chetan (Aitel)	$16, Deepak (BSNL)	$15, Eshan (Jio)	$31, Faiz (BSNAL)	$29,Gopal (Voda)	
Arun ((Airtel)	3	3	X	X	3	3	3	5
Bishwas (Jio)	X	4	3	X	X	4	4	4
Chetan (Airtel)	X	3	3	X	X	3	3	4
Deepak (BSNL)	3	X	X	6	X/4	6	5	5/4
Eshan (Jio)	3	X	X	4	4	4	4	5

Faiz (BSNL)	3	X	X	6	4/X	6	5	4/5
Gopal (Voda)	X	4	3	X	4	5	5	5

The family members who were on the Voda network received calls from Bishwas, Chetan, Eshan, Faiz and Gopal.

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Seven friends - Arun, Bishwas, Chetan, Deepak, Eshan, Faiz and Gopal are subscribers to one of the Cellular Networks - Airtel, Jio, Voda and BSNL.
All members of every family subscribe to the network. Every Network has atleast one customer from among these 7 friends.
All the 7 friends are on a trip to Dubai on a short visit. Every one of made a call to their own families, and in addition atleast two calls to the families of their friends, who were accompanying them on the trip. The roaming charges per call of the 4 networks are : Airtel ($3) ; Jio ($4) ; Voda ($5) and BSNL ($6).
If a subscriber is calling from Network X and the receiver is from Network Y, the charges for the call is equal to Minimum of (Roaming Charges of Network X, Roaming Charges of Network Y).
If they are both from the same network, the charges for the call is equal to the roaming charges of that network.
The additional information are as below:
(i) The total call charges for two friends were $9 and $12. The friend(s) on Jio network called the family members of at least one friend, who was on the Airtel Network.
(ii) Faiz and Gopal talked to the family members of all friends and the average call charges between them was $30. The call charges for Faiz was more than that for Gopal, who is on the Voda network.
(iii) Deepak, who is on the BSNL network spend $16 on the call charges.
(iv) Arun called all the families who were receivers of calls from a friend who is a BSNL subscriber. All the families who received calls from Eshan belonged to different networks.
(v) No family received more than five calls. The family members of Bishwas and Chetan received 4 calls. Chetan had not called Eshan's family.
(vi) The total call charges of Deepak, Eshan and Bishwas were consecutive numbers in the decreasing order.
(vii) Family members of any friend can receive a call from atmost one Airtel subscriber.
(viii) Family members of two friends on the same network must be contacted by different number of friends.

Q.6 [11831809]
Family members of Faiz did not receive calls from

1 ◯ Bishwas and Eshan

2 ◯ Bishwas and Chetan

3 ◯ Bishwas, Chetan and Eshan

4 ◯ Chetan and Eshan

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:

From condition (iii), Deepak, who is on the BSNL network spend $16 on the call charges.

From condition (vi), The total call charges of Deepak, Eshan and Bishwas were $16, $15 and $14.

From condition (ii), Faiz and Gopal talked to the family members of all friends and the average call charges between them was $30. So their total call charges were $60. The call charges for Faiz was more than that for Gopal, who is on the Voda network.

So, total call charges for Faiz were more than $30 and total call charges for Gopal were less than $30. Hence, Faiz is on the BSNL network.

Till now, it is clear that there are 2 on BSNL network and 1 of each i.e., Airtel, Jio and Voda networks.

Based on the above data; the total call charges for Faiz = 3 + 4 + 5 + 6 + 6 = $24.

And; that for Gopal = 3 + 4 + 5 + 5 + 5 = $22.

In order to satisfy; the average call charges for Faiz and Gopal together be equal to $30, there is only one possibility that charges for each of them were $3 and $4 for the remaining two calls.

So, these two calls were made to Airtel and Jio networks.

Therefore, Airtel, Jio, Voda and BSNL networks have 2, 2, 1 and 2 subscribers respectively.

Hence, total call charges for Faiz = 24 + 3 + 4 = $31 and total call charges for Gopal = 22 + 3 + 4 = $29.

Step 2:

From condition (i), The total call charges of $9 can be for Airtel subscriber only in the form of 3 + 3 + 3 (= $9).

The total call charges of $12 will also be applicable to the Airtel subscriber as the Jio subscriber will have to make calls to the family members of at least one friend on the Airtel network.

So, the call charges of Jio subscriber must be $4 + $3 for two mandatory calls i.e., one call to his family members and other to Airtel subscriber family members.

Next; Call charges for Jio subscribers can be either $3 or $4; A total call charges of $12 is not possible.

From condition (iv), The total call charges for Arun is $12 and that for Chetan is $9.

From the given information, Every one of made a call to their own families.

From condition (v), Chetan had not called Eshan's family.

The information can be shown in the following table.

Family Members	Friend						
	$12, Arun (Airtel)	$14, Bishwas (Jio)	$9, Chetan (Aitel)	$16, Deepak (BSNL)	$15, Eshan (Jio)	$31, Faiz (BSNAL)	$29,Gopal (Voda)
Arun ((Airtel)	3					3	3
Bishwas (Jio)		4				4	4
Chetan (Airtel)			3			3	3
Deepak (BSNL)				6		6	5
Eshan (Jio)			X		4	4	4
Faiz (BSNL)						6	5
Gopal (Voda)						5	5

Step 3:

From condition (iv), Arun called all the families who were receivers of calls from a friend who is a BSNL subscriber. All the families who received calls from Eshan belonged to different networks.

The total call charges for Bishwas are $14, which can be as 4 + 3 + 4 + 3. It means he must call to family members of both Airtel subscribers.

From condition (v), The family members of Bishwas and Chetan received 4 calls.

The total call charges for Deepak are $16 as 6 + 4 + 6 only.

Hence, the total information can be shown in the following table.

Family Members	Friend							Family Calls
	$12, Arun (Airtel)	$14, Bishwas (Jio)	$9, Chetan (Aitel)	$16, Deepak (BSNL)	$15, Eshan (Jio)	$31, Faiz (BSNAL)	$29,Gopal (Voda)	
Arun ((Airtel)	3	3	X	X	3	3	3	5
Bishwas (Jio)	X	4	3	X	X	4	4	4
Chetan (Airtel)	X	3	3	X	X	3	3	4
Deepak (BSNL)	3	X	X	6	X/4	6	5	5/4
Eshan (Jio)	3	X	X	4	4	4	4	5

Faiz (BSNL)	3	X	X	6	4/X	6	5	4/5
Gopal (Voda)	X	4	3	X	4	5	5	5

Family members of Faiz did not receive calls from Bishwas and Chetan.

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Seven friends - Arun, Bishwas, Chetan, Deepak, Eshan, Faiz and Gopal are subscribers to one of the Cellular Networks - Airtel, Jio, Voda and BSNL.
All members of every family subscribe to the network. Every Network has atleast one customer from among these 7 friends.
All the 7 friends are on a trip to Dubai on a short visit. Every one of made a call to their own families, and in addition atleast two calls to the families of their friends, who were accompanying them on the trip. The roaming charges per call of the 4 networks are : Airtel ($3) ; Jio ($4) ; Voda ($5) and BSNL ($6).
If a subscriber is calling from Network X and the receiver is from Network Y, the charges for the call is equal to Minimum of (Roaming Charges of Network X, Roaming Charges of Network Y).
If they are both from the same network, the charges for the call is equal to the roaming charges of that network.
The additional information are as below:
(i) The total call charges for two friends were $9 and $12. The friend(s) on Jio network called the family members of at least one friend, who was on the Airtel Network.
(ii) Faiz and Gopal talked to the family members of all friends and the average call charges between them was $30. The call charges for Faiz was more than that for Gopal, who is on the Voda network.
(iii) Deepak, who is on the BSNL network spend $16 on the call charges.
(iv) Arun called all the families who were receivers of calls from a friend who is a BSNL subscriber. All the families who received calls from Eshan belonged to different networks.
(v) No family received more than five calls. The family members of Bishwas and Chetan received 4 calls. Chetan had not called Eshan's family.
(vi) The total call charges of Deepak, Eshan and Bishwas were consecutive numbers in the decreasing order.
(vii) Family members of any friend can receive a call from atmost one Airtel subscriber.
(viii) Family members of two friends on the same network must be contacted by different number of friends.

Q.7 [11831809]
The total call charges (in $) of Arun was

1 ◯ 9

2 ◯ 12

3 ◯ 14

4 ◯ 15

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:

From condition (iii), Deepak, who is on the BSNL network spend $16 on the call charges.

From condition (vi), The total call charges of Deepak, Eshan and Bishwas were $16, $15 and $14.

From condition (ii), Faiz and Gopal talked to the family members of all friends and the average call charges between them was $30. So their total call charges were $60. The call charges for Faiz was more than that for Gopal, who is on the Voda network.

So, total call charges for Faiz were more than $30 and total call charges for Gopal were less than $30. Hence, Faiz is on the BSNL network.

Till now, it is clear that there are 2 on BSNL network and 1 of each i.e., Airtel, Jio and Voda networks.

Based on the above data; the total call charges for Faiz = 3 + 4 + 5 + 6 + 6 = $24.

And; that for Gopal = 3 + 4 + 5 + 5 + 5 = $22.

In order to satisfy; the average call charges for Faiz and Gopal together be equal to $30, there is only one possibility that charges for each of them were $3 and $4 for the remaining two calls.

So, these two calls were made to Airtel and Jio networks.

Therefore, Airtel, Jio, Voda and BSNL networks have 2, 2, 1 and 2 subscribers respectively.

Hence, total call charges for Faiz = 24 + 3 + 4 = $31 and total call charges for Gopal = 22 + 3 + 4 = $29.

Step 2:

From condition (i), The total call charges of $9 can be for Airtel subscriber only in the form of 3 + 3 + 3 (= $9).

The total call charges of $12 will also be applicable to the Airtel subscriber as the Jio subscriber will have to make calls to the family members of at least one friend on the Airtel network.

So, the call charges of Jio subscriber must be $4 + $3 for two mandatory calls i.e., one call to his family members and other to Airtel subscriber family members.

Next; Call charges for Jio subscribers can be either $3 or $4; A total call charges of $12 is not possible.

From condition (iv), The total call charges for Arun is $12 and that for Chetan is $9.

From the given information, Every one of made a call to their own families.

From condition (v), Chetan had not called Eshan's family.

The information can be shown in the following table.

Family Members	Friend						
	$12, Arun (Airtel)	$14, Bishwas (Jio)	$9, Chetan (Aitel)	$16, Deepak (BSNL)	$15, Eshan (Jio)	$31, Faiz (BSNAL)	$29,Gopal (Voda)
Arun ((Airtel)	3					3	3
Bishwas (Jio)		4				4	4
Chetan (Airtel)			3			3	3
Deepak (BSNL)				6		6	5
Eshan (Jio)			X		4	4	4
Faiz (BSNL)						6	5
Gopal (Voda)						5	5

Step 3:

From condition (iv), Arun called all the families who were receivers of calls from a friend who is a BSNL subscriber. All the families who received calls from Eshan belonged to different networks.

The total call charges for Bishwas are $14, which can be as 4 + 3 + 4 + 3. It means he must call to family members of both Airtel subscribers.

From condition (v), The family members of Bishwas and Chetan received 4 calls.

The total call charges for Deepak are $16 as 6 + 4 + 6 only.

Hence, the total information can be shown in the following table.

Family Members	Friend							Family Calls
	$12, Arun (Airtel)	$14, Bishwas (Jio)	$9, Chetan (Aitel)	$16, Deepak (BSNL)	$15, Eshan (Jio)	$31, Faiz (BSNAL)	$29,Gopal (Voda)	
Arun ((Airtel)	3	3	X	X	3	3	3	5
Bishwas (Jio)	X	4	3	X	X	4	4	4
Chetan (Airtel)	X	3	3	X	X	3	3	4
Deepak (BSNL)	3	X	X	6	X/4	6	5	5/4
Eshan (Jio)	3	X	X	4	4	4	4	5

Faiz (BSNL)	3	X	X	6	4/X	6	5	4/5
Gopal (Voda)	X	4	3	X	4	5	5	5

The total call charges for Arun was $12.

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Seven friends - Arun, Bishwas, Chetan, Deepak, Eshan, Faiz and Gopal are subscribers to one of the Cellular Networks - Airtel, Jio, Voda and BSNL.
All members of every family subscribe to the network. Every Network has atleast one customer from among these 7 friends.
All the 7 friends are on a trip to Dubai on a short visit. Every one of made a call to their own families, and in addition atleast two calls to the families of their friends, who were accompanying them on the trip. The roaming charges per call of the 4 networks are : Airtel ($3) ; Jio ($4) ; Voda ($5) and BSNL ($6).
If a subscriber is calling from Network X and the receiver is from Network Y, the charges for the call is equal to Minimum of (Roaming Charges of Network X, Roaming Charges of Network Y).
If they are both from the same network, the charges for the call is equal to the roaming charges of that network.
The additional information are as below:
(i) The total call charges for two friends were $9 and $12. The friend(s) on Jio network called the family members of at least one friend, who was on the Airtel Network.
(ii) Faiz and Gopal talked to the family members of all friends and the average call charges between them was $30. The call charges for Faiz was more than that for Gopal, who is on the Voda network.
(iii) Deepak, who is on the BSNL network spend $16 on the call charges.
(iv) Arun called all the families who were receivers of calls from a friend who is a BSNL subscriber. All the families who received calls from Eshan belonged to different networks.
(v) No family received more than five calls. The family members of Bishwas and Chetan received 4 calls. Chetan had not called Eshan's family.
(vi) The total call charges of Deepak, Eshan and Bishwas were consecutive numbers in the decreasing order.
(vii) Family members of any friend can receive a call from atmost one Airtel subscriber.
(viii) Family members of two friends on the same network must be contacted by different number of friends.

Q.8 [11831809]
How many friend(s) had not made any call a receiver on the Airtel Network?

1 ◯ 1

2 ◯ 2

3 ◯ 3

4 ◯ Either (1) or (2)

Solution:
Correct Answer : 1

Answer key/Solution

Step 1:

From condition (iii), Deepak, who is on the BSNL network spend $16 on the call charges.

From condition (vi), The total call charges of Deepak, Eshan and Bishwas were $16, $15 and $14.

From condition (ii), Faiz and Gopal talked to the family members of all friends and the average call charges between them was $30. So their total call charges were $60. The call charges for Faiz was more than that for Gopal, who is on the Voda network.

So, total call charges for Faiz were more than $30 and total call charges for Gopal were less than $30. Hence, Faiz is on the BSNL network.

Till now, it is clear that there are 2 on BSNL network and 1 of each i.e., Airtel, Jio and Voda networks.

Based on the above data; the total call charges for Faiz = 3 + 4 + 5 + 6 + 6 = $24.

And; that for Gopal = 3 + 4 + 5 + 5 + 5 = $22.

In order to satisfy; the average call charges for Faiz and Gopal together be equal to $30, there is only one possibility that charges for each of them were $3 and $4 for the remaining two calls.

So, these two calls were made to Airtel and Jio networks.

Therefore, Airtel, Jio, Voda and BSNL networks have 2, 2, 1 and 2 subscribers respectively.

Hence, total call charges for Faiz = 24 + 3 + 4 = $31 and total call charges for Gopal = 22 + 3 + 4 = $29.

Step 2:

From condition (i), The total call charges of $9 can be for Airtel subscriber only in the form of 3 + 3 + 3 (= $9).

The total call charges of $12 will also be applicable to the Airtel subscriber as the Jio subscriber will have to make calls to the family members of at least one friend on the Airtel network.

So, the call charges of Jio subscriber must be $4 + $3 for two mandatory calls i.e., one call to his family members and other to Airtel subscriber family members.

Next; Call charges for Jio subscribers can be either $3 or $4; A total call charges of $12 is not possible.

From condition (iv), The total call charges for Arun is $12 and that for Chetan is $9.

From the given information, Every one of made a call to their own families.

From condition (v), Chetan had not called Eshan's family.

The information can be shown in the following table.

Family Members	Friend						
	$12, Arun (Airtel)	$14, Bishwas (Jio)	$9, Chetan (Aitel)	$16, Deepak (BSNL)	$15, Eshan (Jio)	$31, Faiz (BSNAL)	$29,Gopal (Voda)
Arun ((Airtel)	3					3	3
Bishwas (Jio)		4				4	4
Chetan (Airtel)			3			3	3
Deepak (BSNL)				6		6	5
Eshan (Jio)			X		4	4	4
Faiz (BSNL)						6	5
Gopal (Voda)						5	5

Step 3:

From condition (iv), Arun called all the families who were receivers of calls from a friend who is a BSNL subscriber. All the families who received calls from Eshan belonged to different networks.

The total call charges for Bishwas are $14, which can be as 4 + 3 + 4 + 3. It means he must call to family members of both Airtel subscribers.

From condition (v), The family members of Bishwas and Chetan received 4 calls.

The total call charges for Deepak are $16 as 6 + 4 + 6 only.

Hence, the total information can be shown in the following table.

Family Members	Friend							Family Calls
	$12, Arun (Airtel)	$14, Bishwas (Jio)	$9, Chetan (Aitel)	$16, Deepak (BSNL)	$15, Eshan (Jio)	$31, Faiz (BSNAL)	$29,Gopal (Voda)	
Arun ((Airtel)	3	3	X	X	3	3	3	5
Bishwas (Jio)	X	4	3	X	X	4	4	4
Chetan (Airtel)	X	3	3	X	X	3	3	4
Deepak (BSNL)	3	X	X	6	X/4	6	5	5/4
Eshan (Jio)	3	X	X	4	4	4	4	5

Faiz (BSNL)	3	X	X	6	4/X	6	5	4/5
Gopal (Voda)	X	4	3	X	4	5	5	5

Deepak had not made any call a receiver on the Airtel Network.

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Seven friends - Arun, Bishwas, Chetan, Deepak, Eshan, Faiz and Gopal are subscribers to one of the Cellular Networks - Airtel, Jio, Voda and BSNL.
All members of every family subscribe to the network. Every Network has atleast one customer from among these 7 friends.
All the 7 friends are on a trip to Dubai on a short visit. Every one of made a call to their own families, and in addition atleast two calls to the families of their friends, who were accompanying them on the trip. The roaming charges per call of the 4 networks are : Airtel ($3) ; Jio ($4) ; Voda ($5) and BSNL ($6).
If a subscriber is calling from Network X and the receiver is from Network Y, the charges for the call is equal to Minimum of (Roaming Charges of Network X, Roaming Charges of Network Y).
If they are both from the same network, the charges for the call is equal to the roaming charges of that network.
The additional information are as below:
(i) The total call charges for two friends were $9 and $12. The friend(s) on Jio network called the family members of at least one friend, who was on the Airtel Network.
(ii) Faiz and Gopal talked to the family members of all friends and the average call charges between them was $30. The call charges for Faiz was more than that for Gopal, who is on the Voda network.
(iii) Deepak, who is on the BSNL network spend $16 on the call charges.
(iv) Arun called all the families who were receivers of calls from a friend who is a BSNL subscriber. All the families who received calls from Eshan belonged to different networks.
(v) No family received more than five calls. The family members of Bishwas and Chetan received 4 calls. Chetan had not called Eshan's family.
(vi) The total call charges of Deepak, Eshan and Bishwas were consecutive numbers in the decreasing order.
(vii) Family members of any friend can receive a call from atmost one Airtel subscriber.
(viii) Family members of two friends on the same network must be contacted by different number of friends.

Q.9 [11831809]
Which of the following is true about Eshan's calls?

1 ◯ He called Deepak's family.

2 ◯ He called either Deepak's family or Faiz's family.

3 ◯ He called Chetan's family.

4 ◯ He called either Chetan's family or Faiz's family.

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:

From condition (iii), Deepak, who is on the BSNL network spend $16 on the call charges.

From condition (vi), The total call charges of Deepak, Eshan and Bishwas were $16, $15 and $14.

From condition (ii), Faiz and Gopal talked to the family members of all friends and the average call charges between them was $30. So their total call charges were $60. The call charges for Faiz was more than that for Gopal, who is on the Voda network.

So, total call charges for Faiz were more than $30 and total call charges for Gopal were less than $30. Hence, Faiz is on the BSNL network.

Till now, it is clear that there are 2 on BSNL network and 1 of each i.e., Airtel, Jio and Voda networks.

Based on the above data; the total call charges for Faiz = 3 + 4 + 5 + 6 + 6 = $24.

And; that for Gopal = 3 + 4 + 5 + 5 + 5 = $22.

In order to satisfy; the average call charges for Faiz and Gopal together be equal to $30, there is only one possibility that charges for each of them were $3 and $4 for the remaining two calls.

So, these two calls were made to Airtel and Jio networks.

Therefore, Airtel, Jio, Voda and BSNL networks have 2, 2, 1 and 2 subscribers respectively.

Hence, total call charges for Faiz = 24 + 3 + 4 = $31 and total call charges for Gopal = 22 + 3 + 4 = $29.

Step 2:

From condition (i), The total call charges of $9 can be for Airtel subscriber only in the form of 3 + 3 + 3 (= $9).

The total call charges of $12 will also be applicable to the Airtel subscriber as the Jio subscriber will have to make calls to the family members of at least one friend on the Airtel network.

So, the call charges of Jio subscriber must be $4 + $3 for two mandatory calls i.e., one call to his family members and other to Airtel subscriber family members.

Next; Call charges for Jio subscribers can be either $3 or $4; A total call charges of $12 is not possible.

From condition (iv), The total call charges for Arun is $12 and that for Chetan is $9.

From the given information, Every one of made a call to their own families.

From condition (v), Chetan had not called Eshan's family.

The information can be shown in the following table.

Family Members	Friend						
	$12, Arun (Airtel)	$14, Bishwas (Jio)	$9, Chetan (Aitel)	$16, Deepak (BSNL)	$15, Eshan (Jio)	$31, Faiz (BSNAL)	$29,Gopal (Voda)
Arun ((Airtel)	3					3	3
Bishwas (Jio)		4				4	4
Chetan (Airtel)			3			3	3
Deepak (BSNL)				6		6	5
Eshan (Jio)			X		4	4	4
Faiz (BSNL)						6	5
Gopal (Voda)						5	5

Step 3:

From condition (iv), Arun called all the families who were receivers of calls from a friend who is a BSNL subscriber. All the families who received calls from Eshan belonged to different networks.

The total call charges for Bishwas are $14, which can be as 4 + 3 + 4 + 3. It means he must call to family members of both Airtel subscribers.

From condition (v), The family members of Bishwas and Chetan received 4 calls.

The total call charges for Deepak are $16 as 6 + 4 + 6 only.

Hence, the total information can be shown in the following table.

Family Members	Friend							Family Calls
	$12, Arun (Airtel)	$14, Bishwas (Jio)	$9, Chetan (Aitel)	$16, Deepak (BSNL)	$15, Eshan (Jio)	$31, Faiz (BSNAL)	$29,Gopal (Voda)	
Arun ((Airtel)	3	3	X	X	3	3	3	5
Bishwas (Jio)	X	4	3	X	X	4	4	4
Chetan (Airtel)	X	3	3	X	X	3	3	4
Deepak (BSNL)	3	X	X	6	X/4	6	5	5/4
Eshan (Jio)	3	X	X	4	4	4	4	5

Faiz (BSNL)	3	X	X	6	4/X	6	5	4/5
Gopal (Voda)	X	4	3	X	4	5	5	5

The statement given in option (2) is true about Eshan's calls.

Bookmark FeedBack

Directions for questions 5 to 10: Answer the questions on the basis of the information given below.

Seven friends - Arun, Bishwas, Chetan, Deepak, Eshan, Faiz and Gopal are subscribers to one of the Cellular Networks - Airtel, Jio, Voda and BSNL.
All members of every family subscribe to the network. Every Network has atleast one customer from among these 7 friends.
All the 7 friends are on a trip to Dubai on a short visit. Every one of made a call to their own families, and in addition atleast two calls to the families of their friends, who were accompanying them on the trip. The roaming charges per call of the 4 networks are : Airtel ($3) ; Jio ($4) ; Voda ($5) and BSNL ($6).
If a subscriber is calling from Network X and the receiver is from Network Y, the charges for the call is equal to Minimum of (Roaming Charges of Network X, Roaming Charges of Network Y).
If they are both from the same network, the charges for the call is equal to the roaming charges of that network.
The additional information are as below:
(i) The total call charges for two friends were $9 and $12. The friend(s) on Jio network called the family members of at least one friend, who was on the Airtel Network.
(ii) Faiz and Gopal talked to the family members of all friends and the average call charges between them was $30. The call charges for Faiz was more than that for Gopal, who is on the Voda network.
(iii) Deepak, who is on the BSNL network spend $16 on the call charges.
(iv) Arun called all the families who were receivers of calls from a friend who is a BSNL subscriber. All the families who received calls from Eshan belonged to different networks.
(v) No family received more than five calls. The family members of Bishwas and Chetan received 4 calls. Chetan had not called Eshan's family.
(vi) The total call charges of Deepak, Eshan and Bishwas were consecutive numbers in the decreasing order.
(vii) Family members of any friend can receive a call from atmost one Airtel subscriber.
(viii) Family members of two friends on the same network must be contacted by different number of friends.

Q.10 [11831809]
Who had made the minimum number of calls?

1 ◯ Arun

2 ◯ Chetan

3 ◯ Arun and Chetan

4 ◯ Chetan and Deepak

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:

From condition (iii), Deepak, who is on the BSNL network spend $16 on the call charges.

From condition (vi), The total call charges of Deepak, Eshan and Bishwas were $16, $15 and $14.

From condition (ii), Faiz and Gopal talked to the family members of all friends and the average call charges between them was $30. So their total call charges were $60. The call charges for Faiz was more than that for Gopal, who is on the Voda network.

So, total call charges for Faiz were more than $30 and total call charges for Gopal were less than $30. Hence, Faiz is on the BSNL network.

Till now, it is clear that there are 2 on BSNL network and 1 of each i.e., Airtel, Jio and Voda networks.

Based on the above data; the total call charges for Faiz = 3 + 4 + 5 + 6 + 6 = $24.

And; that for Gopal = 3 + 4 + 5 + 5 + 5 = $22.

In order to satisfy; the average call charges for Faiz and Gopal together be equal to $30, there is only one possibility that charges for each of them were $3 and $4 for the remaining two calls.

So, these two calls were made to Airtel and Jio networks.

Therefore, Airtel, Jio, Voda and BSNL networks have 2, 2, 1 and 2 subscribers respectively.

Hence, total call charges for Faiz = 24 + 3 + 4 = $31 and total call charges for Gopal = 22 + 3 + 4 = $29.

Step 2:

From condition (i), The total call charges of $9 can be for Airtel subscriber only in the form of 3 + 3 + 3 (= $9).

The total call charges of $12 will also be applicable to the Airtel subscriber as the Jio subscriber will have to make calls to the family members of at least one friend on the Airtel network.

So, the call charges of Jio subscriber must be $4 + $3 for two mandatory calls i.e., one call to his family members and other to Airtel subscriber family members.

Next; Call charges for Jio subscribers can be either $3 or $4; A total call charges of $12 is not possible.

From condition (iv), The total call charges for Arun is $12 and that for Chetan is $9.

From the given information, Every one of made a call to their own families.

From condition (v), Chetan had not called Eshan's family.

The information can be shown in the following table.

Family Members	Friend						
	$12, Arun (Airtel)	$14, Bishwas (Jio)	$9, Chetan (Aitel)	$16, Deepak (BSNL)	$15, Eshan (Jio)	$31, Faiz (BSNAL)	$29,Gopal (Voda)
Arun ((Airtel)	3					3	3
Bishwas (Jio)		4				4	4
Chetan (Airtel)			3			3	3
Deepak (BSNL)				6		6	5
Eshan (Jio)			X		4	4	4
Faiz (BSNL)						6	5
Gopal (Voda)						5	5

Step 3:

From condition (iv), Arun called all the families who were receivers of calls from a friend who is a BSNL subscriber. All the families who received calls from Eshan belonged to different networks.

The total call charges for Bishwas are $14, which can be as 4 + 3 + 4 + 3. It means he must call to family members of both Airtel subscribers.

From condition (v), The family members of Bishwas and Chetan received 4 calls.

The total call charges for Deepak are $16 as 6 + 4 + 6 only.

Hence, the total information can be shown in the following table.

Family Members	Friend							Family Calls
	$12, Arun (Airtel)	$14, Bishwas (Jio)	$9, Chetan (Aitel)	$16, Deepak (BSNL)	$15, Eshan (Jio)	$31, Faiz (BSNAL)	$29,Gopal (Voda)	
Arun ((Airtel)	3	3	X	X	3	3	3	5
Bishwas (Jio)	X	4	3	X	X	4	4	4
Chetan (Airtel)	X	3	3	X	X	3	3	4
Deepak (BSNL)	3	X	X	6	X/4	6	5	5/4
Eshan (Jio)	3	X	X	4	4	4	4	5

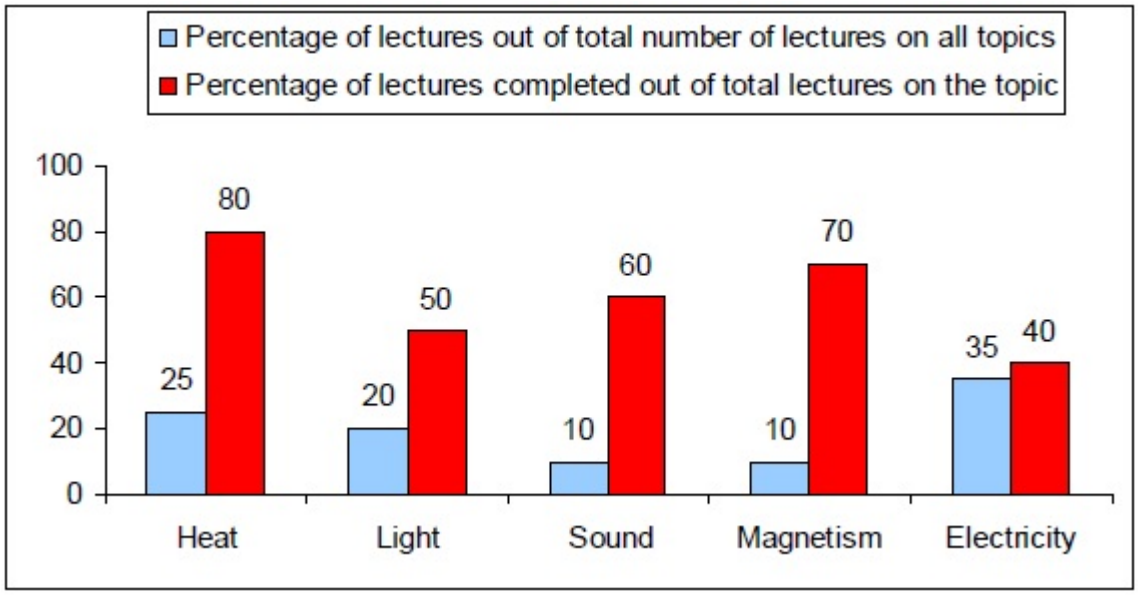
Faiz (BSNL)	3	X	X	6	4/X	6	5	4/5
Gopal (Voda)	X	4	3	X	4	5	5	5

Chetan and Deepak had made the minimum number of calls.

Bookmark FeedBack

Directions for questions 11 to 15: Answer the questions on the basis of the information given below.

Tyagi is a student, who has enrolled in an online Physics course that has a series of lectures on five different topics - Heat, Light, Sound, Magnetism and Electricity. The lectures are recorded videos and the total number of lectures on each topic may or may not be equal but the duration of each lecture is 30 minutes. A student is free to pause a lecture in between and view it later on.
The bar graph given below shows the percentage distribution of the number of lectures on each topic out of the total number of lectures and the percentage of lectures out of the total lectures on each topic that Tyagi has finished viewing.



Further, it is also known that:
(i) The total number of lectures already viewed by Tyagi are 114.
(ii) The time spent by Tyagi each day on viewing the lectures on Heat, Light, Sound, Magnetism and Electricity are 50, 45, 35, 40 and 30 minutes respectively.
(iii) Lectures on each topic are ordered from 1, 2, 3, …, to n.
(iv) Tyagi views the lectures of each topic in order (from 1st to last), that is he watches the (n + 1)th lecture only after completing the nth lecture.

Q.11 [11831809]
Find the number of days required by Tyagi to complete all the remaining lectures on Heat.

1 ◯ 5

2 ◯ 6

3 ◯ 8

4 ◯ 7.5

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:
Let the total number of lectures on all topics be x.

Topic	No. of lectures	No. of lectures completed
Heat	0.25x	0.80 × 0.25x = 0.2x
Light	0.20x	0.50 × 0.2x = 0.1x
Sound	0,10x	0.60 × 0.1x = 0.06x
Magnetism	0.10x	0.70 × 0.1x = 0.07x
Electricity	0.35x	0.40 × 0.35x = 0.14x

From condition (i), (0.2 + 0.1 + 0.06 + 0.07 + 0.14)x = 114 ⇒ x = 200.

Step 2:

Topic	No. of lectures	No. of lectures completed
Heat	50	40
Light	40	20
Sound	20	12
Magnetism	20	14
Electricity	70	28
Total	200	114

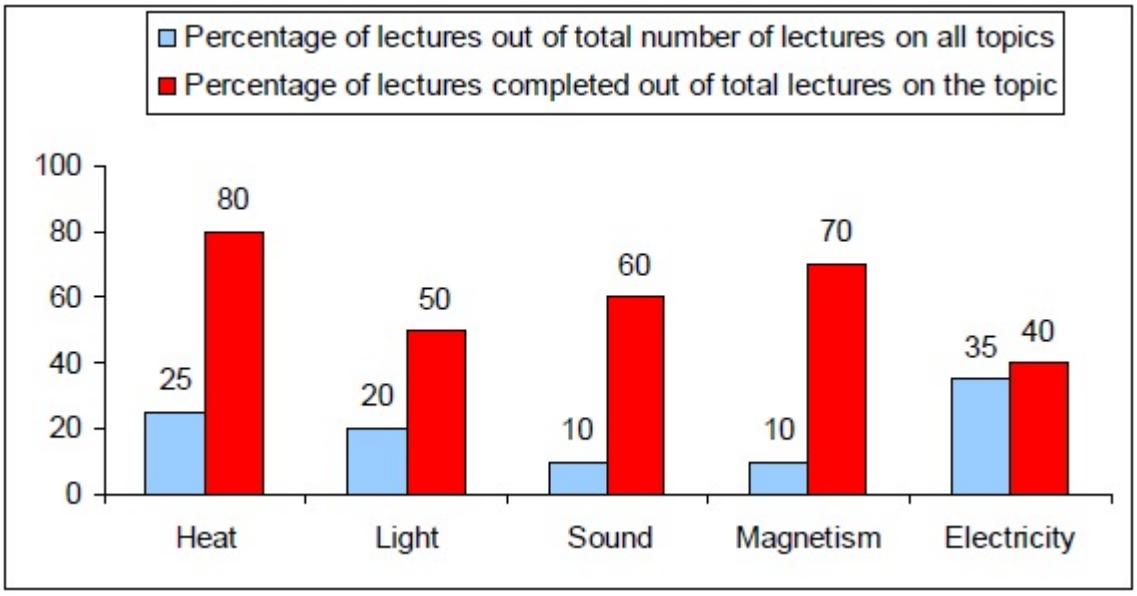
The time spent by Tyagi per day on lectures of Heat, Light, Sound, Magnetism and Electricity are 50, 45, 35, 40 and 30 minutes respectively. Each recorded lecture is of 30 minute duration. On any particular day:
After viewing the first lecture on Heat, Tyagi has 20 minutes to view the second lecture.
After viewing the first lecture on Light Tyagi has 15 minutes to view the second lecture.
After viewing the first lecture on Sound Tyagi has 5 minutes to view the second lecture.
After viewing the first lecture on Magnetism Tyagi has 10 minutes to view the second lecture.
After viewing the first lecture on Electricity Tyagi has no time to view the second lecture.

Total number of lectures on Heat = 50
Number of lectures completed = 40
Time taken to finish 40 lectures = 40 × 30 = 1200 minutes
Time spent everyday = 50 minutes
Hence, number of days spent viewing 40 lectures = 1200/50 = 24 days
(There is no remainder, so the 41st lecture is viewed on a fresh day)
Time taken to complete the remaining 10 lectures = 10 × 30 = 300 minutes
Hence, number of days required = 300/50 = 6 days.

Bookmark FeedBack

Directions for questions 11 to 15: Answer the questions on the basis of the information given below.

Tyagi is a student, who has enrolled in an online Physics course that has a series of lectures on five different topics - Heat, Light, Sound, Magnetism and Electricity. The lectures are recorded videos and the total number of lectures on each topic may or may not be equal but the duration of each lecture is 30 minutes. A student is free to pause a lecture in between and view it later on.
The bar graph given below shows the percentage distribution of the number of lectures on each topic out of the total number of lectures and the percentage of lectures out of the total lectures on each topic that Tyagi has finished viewing.



Further, it is also known that:
(i) The total number of lectures already viewed by Tyagi are 114.
(ii) The time spent by Tyagi each day on viewing the lectures on Heat, Light, Sound, Magnetism and Electricity are 50, 45, 35, 40 and 30 minutes respectively.
(iii) Lectures on each topic are ordered from 1, 2, 3, …, to n.
(iv) Tyagi views the lectures of each topic in order (from 1st to last), that is he watches the (n + 1)th lecture only after completing the nth lecture.

Q.12 [11831809]
Which lecture of Light is Tyagi watching at the end of Day 5?

1 ◯ Lecture 9

2 ◯ Lecture 7

3 ◯ Lecture 6

4 ◯ Lecture 8

Solution:
Correct Answer : 4

Answer key/Solution

Step 1:
Let the total number of lectures on all topics be x.

Topic	No. of lectures	No. of lectures completed
Heat	0.25x	0.80 × 0.25x = 0.2x
Light	0.20x	0.50 × 0.2x = 0.1x
Sound	0,10x	0.60 × 0.1x = 0.06x
Magnetism	0.10x	0.70 × 0.1x = 0.07x
Electricity	0.35x	0.40 × 0.35x = 0.14x

From condition (i), (0.2 + 0.1 + 0.06 + 0.07 + 0.14)x = 114 ⇒ x = 200.

Step 2:

Topic	No. of lectures	No. of lectures completed
Heat	50	40
Light	40	20
Sound	20	12
Magnetism	20	14
Electricity	70	28
Total	200	114

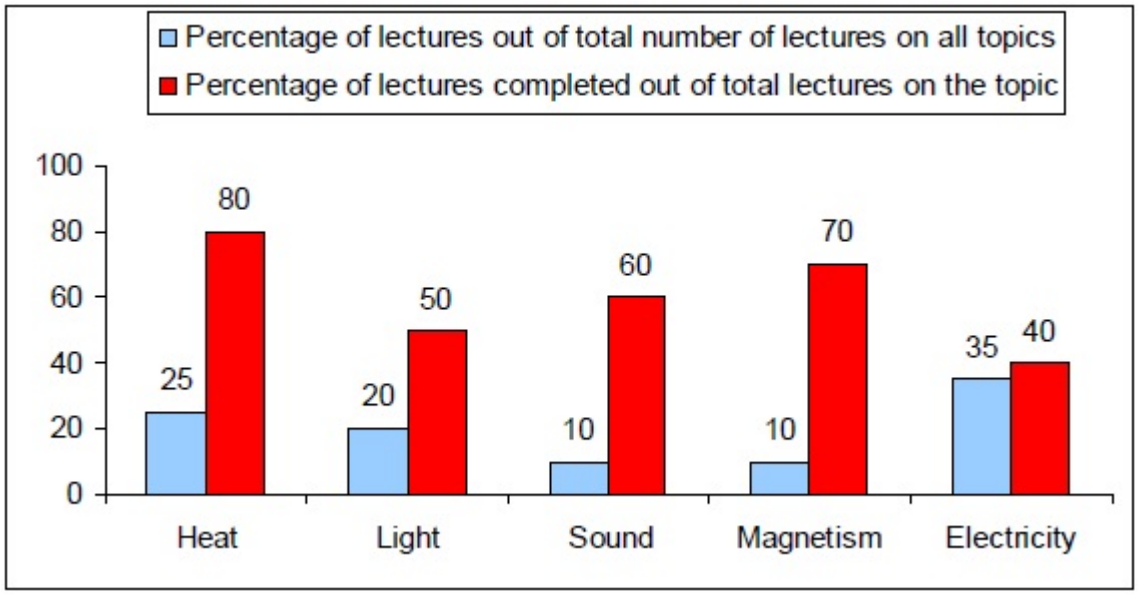
The time spent by Tyagi per day on lectures of Heat, Light, Sound, Magnetism and Electricity are 50, 45, 35, 40 and 30 minutes respectively. Each recorded lecture is of 30 minute duration. On any particular day:
After viewing the first lecture on Heat, Tyagi has 20 minutes to view the second lecture.
After viewing the first lecture on Light Tyagi has 15 minutes to view the second lecture.
After viewing the first lecture on Sound Tyagi has 5 minutes to view the second lecture.
After viewing the first lecture on Magnetism Tyagi has 10 minutes to view the second lecture.
After viewing the first lecture on Electricity Tyagi has no time to view the second lecture.

Time spent by Tyagi per day on each lecture of Light is 45 minutes. So everyday he views 1 full and one half of a lecture. In 5 days he views 7.5 lectures. This means that at the end of his viewing time on Day 5 Tyagi is watching Lecture 8 of Light.

Bookmark FeedBack

Directions for questions 11 to 15: Answer the questions on the basis of the information given below.

Tyagi is a student, who has enrolled in an online Physics course that has a series of lectures on five different topics - Heat, Light, Sound, Magnetism and Electricity. The lectures are recorded videos and the total number of lectures on each topic may or may not be equal but the duration of each lecture is 30 minutes. A student is free to pause a lecture in between and view it later on.
The bar graph given below shows the percentage distribution of the number of lectures on each topic out of the total number of lectures and the percentage of lectures out of the total lectures on each topic that Tyagi has finished viewing.



Further, it is also known that:
(i) The total number of lectures already viewed by Tyagi are 114.
(ii) The time spent by Tyagi each day on viewing the lectures on Heat, Light, Sound, Magnetism and Electricity are 50, 45, 35, 40 and 30 minutes respectively.
(iii) Lectures on each topic are ordered from 1, 2, 3, …, to n.
(iv) Tyagi views the lectures of each topic in order (from 1st to last), that is he watches the (n + 1)th lecture only after completing the nth lecture.

Q.13 [11831809]
If the time spent by Tyagi each day on viewing lectures of Electricity was the same as that spent on lectures of Sound, then how many days earlier than the scheduled time would Tyagi complete the topic of Electricity?

1 ◯ 8

2 ◯ 9

3 ◯ 10

4 ◯ 11

Solution:
Correct Answer : 3

Answer key/Solution

Step 1:
Let the total number of lectures on all topics be x.

Topic	No. of lectures	No. of lectures completed
Heat	0.25x	0.80 × 0.25x = 0.2x
Light	0.20x	0.50 × 0.2x = 0.1x
Sound	0,10x	0.60 × 0.1x = 0.06x
Magnetism	0.10x	0.70 × 0.1x = 0.07x
Electricity	0.35x	0.40 × 0.35x = 0.14x

From condition (i), (0.2 + 0.1 + 0.06 + 0.07 + 0.14)x = 114 ⇒ x = 200.

Step 2:

Topic	No. of lectures	No. of lectures completed
Heat	50	40
Light	40	20
Sound	20	12
Magnetism	20	14
Electricity	70	28
Total	200	114

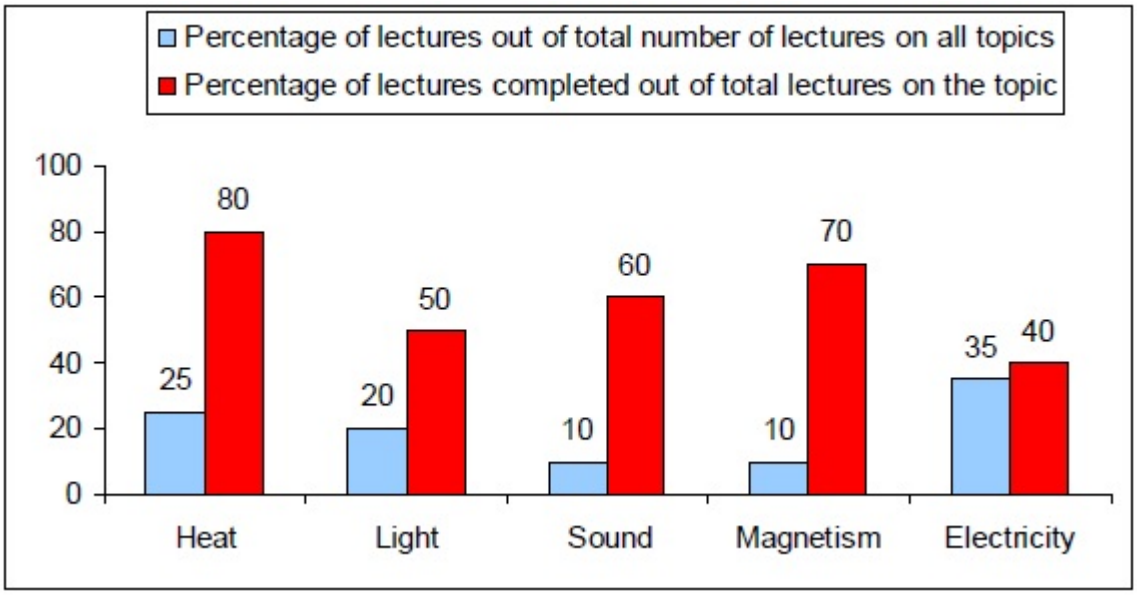
The time spent by Tyagi per day on lectures of Heat, Light, Sound, Magnetism and Electricity are 50, 45, 35, 40 and 30 minutes respectively. Each recorded lecture is of 30 minute duration. On any particular day:
After viewing the first lecture on Heat, Tyagi has 20 minutes to view the second lecture.
After viewing the first lecture on Light Tyagi has 15 minutes to view the second lecture.
After viewing the first lecture on Sound Tyagi has 5 minutes to view the second lecture.
After viewing the first lecture on Magnetism Tyagi has 10 minutes to view the second lecture.
After viewing the first lecture on Electricity Tyagi has no time to view the second lecture.

Tyagi spends 35 minutes on lectures of Sound.
Number of days taken to complete all the lectures of Electricity at 35 minutes per day
= (70 × 30)/35 = 60 days
Hence, Tyagi would complete the topic 10 days before schedule.

Bookmark FeedBack

Directions for questions 11 to 15: Answer the questions on the basis of the information given below.

Tyagi is a student, who has enrolled in an online Physics course that has a series of lectures on five different topics - Heat, Light, Sound, Magnetism and Electricity. The lectures are recorded videos and the total number of lectures on each topic may or may not be equal but the duration of each lecture is 30 minutes. A student is free to pause a lecture in between and view it later on.
The bar graph given below shows the percentage distribution of the number of lectures on each topic out of the total number of lectures and the percentage of lectures out of the total lectures on each topic that Tyagi has finished viewing.



Further, it is also known that:
(i) The total number of lectures already viewed by Tyagi are 114.
(ii) The time spent by Tyagi each day on viewing the lectures on Heat, Light, Sound, Magnetism and Electricity are 50, 45, 35, 40 and 30 minutes respectively.
(iii) Lectures on each topic are ordered from 1, 2, 3, …, to n.
(iv) Tyagi views the lectures of each topic in order (from 1st to last), that is he watches the (n + 1)th lecture only after completing the nth lecture.

Q.14 [11831809]
Find the percentage out of the total number of lectures on Magnetism that are remaining to be viewed by Tyagi after Day 9?

1 ◯ 40%

2 ◯ 25%

3 ◯ 20%

4 ◯ 30%

Solution:
Correct Answer : 1

Answer key/Solution

Step 1:
Let the total number of lectures on all topics be x.

Topic	No. of lectures	No. of lectures completed
Heat	0.25x	0.80 × 0.25x = 0.2x
Light	0.20x	0.50 × 0.2x = 0.1x
Sound	0,10x	0.60 × 0.1x = 0.06x
Magnetism	0.10x	0.70 × 0.1x = 0.07x
Electricity	0.35x	0.40 × 0.35x = 0.14x

From condition (i), $(0.2 + 0.1 + 0.06 + 0.07 + 0.14)x = 114 \Rightarrow x = 200$.

Step 2:

Topic	No. of lectures	No. of lectures completed
Heat	50	40
Light	40	20
Sound	20	12
Magnetism	20	14
Electricity	70	28
Total	200	114

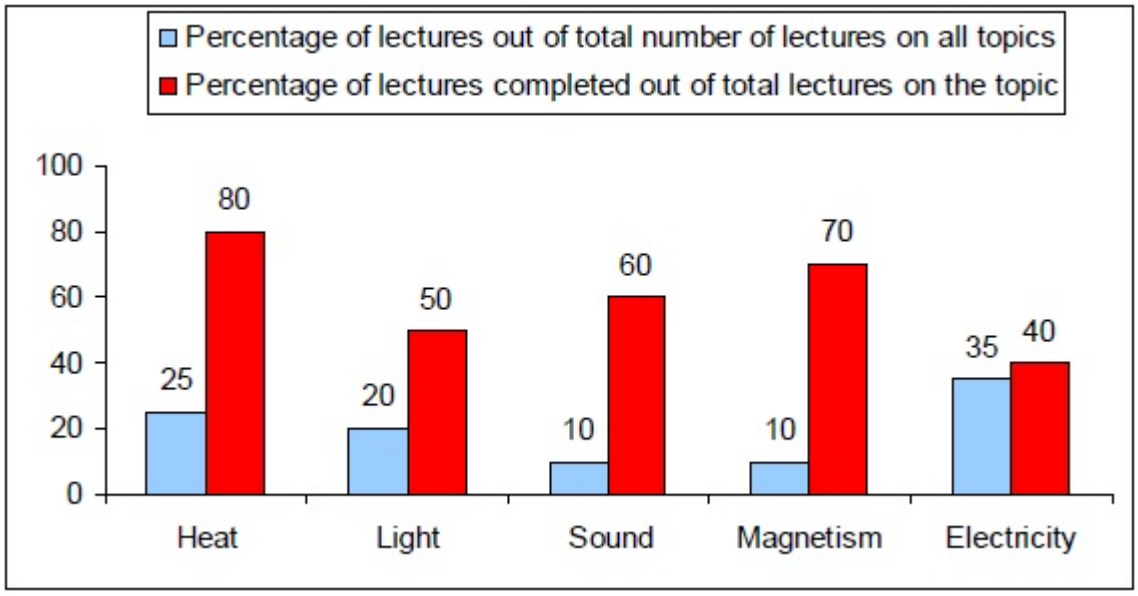
The time spent by Tyagi per day on lectures of Heat, Light, Sound, Magnetism and Electricity are 50, 45, 35, 40 and 30 minutes respectively. Each recorded lecture is of 30 minute duration. On any particular day:
After viewing the first lecture on Heat, Tyagi has 20 minutes to view the second lecture.
After viewing the first lecture on Light Tyagi has 15 minutes to view the second lecture.
After viewing the first lecture on Sound Tyagi has 5 minutes to view the second lecture.
After viewing the first lecture on Magnetism Tyagi has 10 minutes to view the second lecture.
After viewing the first lecture on Electricity Tyagi has no time to view the second lecture.

Tyagi watches lectures on Magnetism for 40 minutes everyday.
On day 1 he watches 30 minutes of lecture 1 and 10 minutes of lecture 2.
On day 2 he watches 20 minutes of lecture 2 and 20 minutes of lecture 3.
On day 3 he watches 10 minutes of lecture 3 and 30 minutes of lecture 4.
Similarly, in every 3 days he completes 4 lectures.
So on the 9th day he completes 12 lectures of Magnetism.
There are a total of 20 lectures on Magnetism and 8 are remaining.
Hence, required percentage = 8/20 × 100 = 40%.

Bookmark FeedBack

Directions for questions 11 to 15: Answer the questions on the basis of the information given below.

Tyagi is a student, who has enrolled in an online Physics course that has a series of lectures on five different topics - Heat, Light, Sound, Magnetism and Electricity. The lectures are recorded videos and the total number of lectures on each topic may or may not be equal but the duration of each lecture is 30 minutes. A student is free to pause a lecture in between and view it later on.
The bar graph given below shows the percentage distribution of the number of lectures on each topic out of the total number of lectures and the percentage of lectures out of the total lectures on each topic that Tyagi has finished viewing.



Further, it is also known that:
(i) The total number of lectures already viewed by Tyagi are 114.
(ii) The time spent by Tyagi each day on viewing the lectures on Heat, Light, Sound, Magnetism and Electricity are 50, 45, 35, 40 and 30 minutes respectively.
(iii) Lectures on each topic are ordered from 1, 2, 3, …, to n.
(iv) Tyagi views the lectures of each topic in order (from 1st to last), that is he watches the (n + 1)th lecture only after completing the nth lecture.

Q.15 [11831809]
What is the total number of lectures that Tyagi has finished (completed) viewing at the end of Day 4?

1 ◯ 24

2 ◯ 25

3 ◯ 28

4 ◯ 30

Solution:
Correct Answer : 2

Answer key/Solution

Step 1:
Let the total number of lectures on all topics be x.

Topic	No. of lectures	No. of lectures completed
Heat	0.25x	0.80 × 0.25x = 0.2x
Light	0.20x	0.50 × 0.2x = 0.1x
Sound	0,10x	0.60 × 0.1x = 0.06x
Magnetism	0.10x	0.70 × 0.1x = 0.07x
Electricity	0.35x	0.40 × 0.35x = 0.14x

From condition (i), (0.2 + 0.1 + 0.06 + 0.07 + 0.14)x = 114 ⇒ x = 200.

Step 2:

Topic	No. of lectures	No. of lectures completed
Heat	50	40
Light	40	20
Sound	20	12
Magnetism	20	14
Electricity	70	28
Total	200	114

The time spent by Tyagi per day on lectures of Heat, Light, Sound, Magnetism and Electricity are 50, 45, 35, 40 and 30 minutes respectively. Each recorded lecture is of 30 minute duration. On any particular day:
After viewing the first lecture on Heat, Tyagi has 20 minutes to view the second lecture.
After viewing the first lecture on Light Tyagi has 15 minutes to view the second lecture.
After viewing the first lecture on Sound Tyagi has 5 minutes to view the second lecture.
After viewing the first lecture on Magnetism Tyagi has 10 minutes to view the second lecture.
After viewing the first lecture on Electricity Tyagi has no time to view the second lecture.

Note that each lecture is 30 minutes long and the total time spent on each topic is different.
Let us find the number of lectures viewed in each topic:
Heat = (50 × 4)/30 = 6.67
Light = (45 × 4)/30 = 6
Sound = (35 × 4)/30 = 4.67
Magnetism = (40 × 4)/30 = 5.33
Electricity = (30 × 4)/30 = 4
Hence, number of lectures finished by Tyagi = 6 + 6 + 4 + 5 + 4 = 25.

Bookmark FeedBack

Directions for questions 16 to 20: Answer the questions on the basis of the information given below.

A Ticketing Program is needed to allow a Wildlife Park to sell tickets. A booking consists of one or more tickets for the same day(s) and can be made up to a week in advance. A booking can be made for a visit of one day or two consecutive days. A booking will be valid for the day(s) chosen only. If you wish to change from one ticket type to another ticket type there is a $4 charge imposed as penalty. The additional payments (if any) due to pricing change of the new ticket type will have to be borne by the customer.

Ticket Type	Cost for 1 day	Cost for 2 days
One Adult	$20	$30
One Child	$12	$18
One Senior	$16	$24
Family Ticket (Allowed upto 2 Adults/Seniors, and 3 Children)	$60	$90
Group of 6 (price per person)	$15	$22.50

Q.16 [11831809]

Raju's family of 10 (2 Seniors, 4 Adults, 4 Children) plan to visit the Wildlife Park. What is the best deal (in $) for 2 days that Raju can get?

Answer key/Solution

Solution:
Correct Answer : 207
Case 1: Cost of 2 Seniors, 4 Adults and 4 Children
= 2 × 24 + 4 × 30 + 4 × 18 = $240
Case 2: Cost of 2 Seniors, 4 Adults and 4 Children
= Family ticket of 2 Adults and 3 Children + 2 Seniors + 2 Adults + 1 Child
= 90 + 2 × 24 + 2 × 30 + 18 = $216
Case 3: Cost of 2 Seniors, 4 Adults and 4 Children
= Group of 6 (2 Seniors, 4 Adults) + 4 Children
= 6 × 22.5 + 4 × 18 = $207
Hence, the cheapest deal is $207.

Bookmark FeedBack

Directions for questions 16 to 20: Answer the questions on the basis of the information given below.

A ticketing program is needed to allow a Wildlife Park to sell tickets. A booking consists of one or more tickets for the same day(s) and can be made up to a week in advance. A booking can be made for a visit of one day or two consecutive days. A booking will be valid for the day(s) chosen only. If you wish to change from one ticket type to another ticket type there is a $4 charge imposed as penalty. The additional payments (if any) due to pricing change of the new ticket type will have to be borne by the customer.

Ticket Type	Cost for 1 day	Cost for 2 days
One Adult	$20	$30
One Child	$12	$18
One Senior	$16	$24
Family Ticket (Allowed upto 2 Adults/Seniors, and 3 Children)	$60	$90
Group of 6 (price per person)	$15	$22.50

Q.17 [11831809]

If the best deal of visiting the Wildlife Part for one day was $ 120 for a group of 10, find the maximum number of Children in the group, if at least one among them is not a child?

1 ○ 5

2 ○ 6

3 ○ 7

4 ○ 8

Answer key/Solution

Solution:
Correct Answer : 4
Let us do the question using the options.
Consider there are 8 children in the family.
Let there be 2 Adults, other than 8 Children.
Family ticket (2 Adults, 3 Children) = $60
Ticket price of 5 Children = 12 × 5 = $60
Total price = $120
Similarly, if there are 2 Seniors or 1 Adult and 1 Senior instead of 2 Adults, the value will be the same.
Let us consider that there are 9 Children, since 10 is not possible as atleast one member is not a child.
Let the other person be Adult. The best price of the ticket will not amount to $120.
Therefore, 9 Children are not possible.
Hence, the maximum number of Children in the group is 8.

Bookmark FeedBack

Directions for questions 16 to 20: Answer the questions on the basis of the information given below.

A ticketing program is needed to allow a Wildlife Park to sell tickets. A booking consists of one or more tickets for the same day(s) and can be made up to a week in advance. A booking can be made for a visit of one day or two consecutive days. A booking will be valid for the day(s) chosen only. If you wish to change from one ticket type to another ticket type there is a $4 charge imposed as penalty. The additional payments (if any) due to pricing change of the new ticket type will have to be borne by the customer.

Ticket Type	Cost for 1 day	Cost for 2 days
One Adult	$20	$30
One Child	$12	$18
One Senior	$16	$24
Family Ticket (Allowed upto 2 Adults/Seniors, and 3 Children)	$60	$90
Group of 6 (price per person)	$15	$22.50

Q.18 [11831809]

Ravi's family of 4 (2 Adults and 2 Children) booked a ticket for one day. However, his brother's family of 3 (2 Adults and 1 Child) and his Parents (2 Seniors) also wanted to tag along. What is the most optimum cost (in $) at which the trip can now be made? (inclusive of any changes of made in the ticket type)

1 ◯ 126

2 ◯ 130

3 ◯ 158

4 ◯ 172

Answer key/Solution

Solution:
Correct Answer : 2
Initially, Ravi must have bought the family tickets for 2 Adults and 2 Children = $60
But since 2 Adults, 1 Child and 2 Seniors are also joining the trip now, therefore, they can now take a group ticket for
4 Adults and 2 Seniors and 3 tickets individually for Children.
Hence, total ticket cost = 6 × 15 + 3 × 12 + 4 (Penalty) = $130.

Bookmark FeedBack

Directions for questions 16 to 20: Answer the questions on the basis of the information given below.

A ticketing program is needed to allow a Wildlife Park to sell tickets. A booking consists of one or more tickets for the same day(s) and can be made up to a week in advance. A booking can be made for a visit of one day or two consecutive days. A booking will be valid for the day(s) chosen only. If you wish to change from one ticket type to another ticket type there is a $4 charge imposed as penalty. The additional payments (if any) due to pricing change of the new ticket type will have to be borne by the customer.

Ticket Type	Cost for 1 day	Cost for 2 days
One Adult	$20	$30
One Child	$12	$18
One Senior	$16	$24
Family Ticket (Allowed upto 2 Adults/Seniors, and 3 Children)	$60	$90
Group of 6 (price per person)	$15	$22.50

Q.19 [11831809]

A group of N people (including Seniors, Adults, and atmost 3 Children) found that the best ticket for a one day visit was to buy One Group of 6 and One Family Ticket, because of which they saved $34 compared to buying individual tickets. What is the number of Adults in this group?

Answer key/Solution

Solution:
Correct Answer : 5
The value of tickets for one group of 6 and one family ticket is 90 + 60 = $150
They saved $34, which means individual tickets would have cost $184.
One group of 6 and one family ticket would mean that the group has either 2 or 3 Children, since, the best tickets are bought. Also, there will be no children in group ticket then.
Case 1: If there are 2 children included in the family ticket.
Then, N = 10.
Individual tickets for 2 Children would cost $24. Out of the remaining $160 let there be x Adults and y Seniors.
20x + 16y = 160 and x + y = 8
Solving we get, x = 8 and y = 0
This case is not possible.

Case 2: If there are 3 children included in the family ticket.
Then, N = 11.
Individual tickets for 3 Children would cost $36. Out of the remaining $148 let there be x Adults and y Seniors.
20x + 16y = 148 and x + y = 8
Solving we get, x = 5 and y = 3
This case is possible.
Hence, there are 5 Adults in the group.

Bookmark FeedBack

Directions for questions 16 to 20: Answer the questions on the basis of the information given below.

A program is needed to allow a Wildlife Park to sell tickets. A booking consists of one or more tickets for the same day(s) and can be made up to a week in advance. A booking can be made for a visit of one day or two consecutive days. A booking will be valid for the day(s) chosen only. If you wish to change from one ticket type to another ticket type there is a $4 charge imposed as penalty. The additional payments (if any) due to pricing change of the new ticket type will have to be borne by the customer.

Ticket Type	Cost for 1 day	Cost for 2 days
One Adult	$20	$30
One Child	$12	$18
One Senior	$16	$24
Family Ticket (Allowed upto 2 Adults/Seniors, and 3 Children)	$60	$90
Group of 6 (price per person)	$15	$22.50

Q.20 [11831809]

Raju's family of 8 (2 Adults, 4 Seniors, 2 Children) had planned for a 2 day trip. But due to some emergency, Raju and his Father (One Adult, One Senior) had to get back to their hometown at the end of day one. So they bought the tickets accordingly. The rest of the family enjoyed their day two in the park. What is the best deal cost (in $) for the trip?

1 ◯ 159

2 ◯ 204

3 ◯ 212

4 ◯ 216

Answer key/Solution

Solution:
Correct Answer : 1
Given that one Adult and one Senior member go for a one day trip and the remaining family (6 members) enjoys a two day trip. A group of 6 is not defined, so anybody can be a part of the group. Their best choice would be to purchase a group ticket for 2 days and 2 tickets for the 2 Children for one day.
On the first day (2 Adults + 4 Seniors) are a part of the group whereas on the second day (1 Adult + 3 Seniors + 2 Children) are part of the group of 6.
Buying tickets for a group of 6 would cost 22.5 × 6 = $135
whereas buying individual tickets for 2 Children = 12 × 2 = $24
Hence, the total cost would add up to = 135 + 24 = $159.

Bookmark FeedBack